WEEK 4 · LESSON 24 OF 24 · HANDOUT

# Capstone Presentation, Final Exam and Certificate

Study time: **about 90 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=9B02MzWwkSo — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50x 2026: Artificial Intelligence — full lecture — https://www.youtube.com/watch?v=-9bo8HlSxwQ

🎓 **Go deeper:** SciPy 2023: Responsible AI in Practice (Rumman Chowdhury) — https://www.youtube.com/watch?v=dQHjntrqrFQ

**Your goal in this lesson:** Present your capstone clearly and honestly, check your submission is complete, prepare for the final exam, and plan your next steps after the WAMUGII AI Program.

## The capstone report

Write 6–10 pages, in simple English, with these sections:

1. **Summary** — the problem, what you built, the result in one number, and your recommendation.
2. **Problem and users** — who has the problem and how it is solved today.
3. **Data** — source, permission, datasheet summary, cleaning and validation.
4. **Approach** — baselines, models tried (from your run log), the chosen model and why.
5. **Results** — metrics on unseen data, compared with the baseline.
6. **Responsible AI** — fairness by group, explanations, privacy, security, human oversight.
7. **Deployment** — API, tests, Docker, monitoring plan and alert rules.
8. **Governance** — risk register, impact assessment, model card, owners.
9. **Limitations and next steps** — what does not work yet, and what you would do with more time.

## The ten-minute presentation

| Minutes | Content |
|---|---|
| 1 | The problem, told as a short story about one user |
| 2 | Data and approach — one slide, no code |
| 2 | **Live demo**: one good case, one hard case, one refused input |
| 2 | Results vs baseline, and fairness by group |
| 2 | Risks, controls and limitations — honestly |
| 1 | What you recommend and what you need |

Tips: practise the demo twice, have screenshots ready in case the internet fails, and answer "I don't know yet — here is how I would find out" when you do not know.

## Check your submission

```
import pathlib

REQUIRED = {
    "README.md": "how to install, train, test and run",
    "requirements.txt": "pinned library versions",
    "Dockerfile": "packaging",
    "DATASHEET.md": "dataset documentation",
    "MODEL_CARD.md": "model documentation",
    "RISK_REGISTER.csv": "risks, controls and owners",
    "tests": "automated tests (folder)",
    "report.pdf": "the written report",
}

def check_submission(folder):
    root = pathlib.Path(folder)
    missing = [name for name in REQUIRED if not (root / name).exists()]
    for name, why in REQUIRED.items():
        print(("[x] " if name not in missing else "[ ] ") + f"{name:18s} {why}")
    return missing

demo = pathlib.Path("my_capstone"); (demo / "tests").mkdir(parents=True, exist_ok=True)
for name in ["README.md", "requirements.txt", "Dockerfile", "MODEL_CARD.md"]:
    (demo / name).write_text("draft\n")
missing = check_submission(demo)
print("\nREADY TO SUBMIT" if not missing else f"\nstill missing: {', '.join(missing)}")
```

Run `check_submission` on your own project folder before you submit.

## Marking rubric

| Area | Weight | Excellent work |
|---|---|---|
| Problem and value | 10% | Real users, clear success measure, beats a baseline |
| Technical quality | 25% | Clean data, sound modelling, reproducible, tested |
| Deployment | 20% | Working API, Docker, CI, monitoring plan |
| Responsible AI | 25% | Fairness, privacy, security, oversight — measured and honest |
| Documentation | 10% | Datasheet, model card, risk register, README |
| Communication | 10% | Clear report and confident demo |

## Final exam and certificate

When you have finished all 24 lessons and their quizzes, take the **final exam** for this course. It covers deployment, MLOps, fairness, explainability, privacy, security, governance and the capstone. With a passing score you receive your **WAMUGII certificate in Responsible AI & Deployment**.

Completing all seven courses of the WAMUGII AI Program means you can now:

1. Use AI tools well and explain how AI works (Courses 1–2).
2. Program in Python and use the maths behind AI (Course 3).
3. Build and evaluate machine-learning and deep-learning models (Courses 4–5).
4. Build LLM and generative-AI applications (Course 6).

5. Deploy, monitor and govern AI systems responsibly (Course 7).

## What next?

- **Build in public:** put your capstone on GitHub with a clear README and a short demo video.
- **Contribute:** open-source projects, Kaggle competitions, local hackathons and AI communities in Kigali.
- **Specialise:** MLOps engineering, data science, NLP for African languages, computer vision for agriculture or health, AI policy and governance.
- **Keep learning responsibly:** every new model you build deserves the same questions — does it work, for whom, what can go wrong, and who is accountable?

**Remember this:** A great capstone is useful, working, honest and well documented — present it clearly, pass the final exam, and keep building AI that serves people.

Name: _____ Date: _____

## Activity

### Task 1

Finish your capstone, run `check_submission` on your folder until it says READY TO SUBMIT, rehearse your ten-minute presentation, and share your project link with your instructor in the lesson Q&A. Congratulations on completing the WAMUGII AI Program!

## Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What should the first minute of the presentation cover?
   - A. The code
   - B. The library versions
   - C. Your hobbies
   - D. The problem, told as a short story about one user
2. What should you prepare in case the live demo fails?
   - A. An excuse
   - B. Screenshots of the demo
   - C. A different project
   - D. Nothing
3. Which area carries the highest weight in the rubric, together with technical quality?
   - A. Documentation
   - B. Responsible AI
   - C. Communication
   - D. Problem and value
4. What should you say when you do not know an answer?
   - A. Make something up
   - B. 'I don't know yet — here is how I would find out'
   - C. Nothing
   - D. Change the subject
5. What does check_submission do?
   - A. Lists required files and shows which are missing
   - B. Trains the model
   - C. Uploads the report
   - D. Grades the project
6. Which question should you ask about every model you build?
   - A. Can it run without data?
   - B. Is it the biggest?
   - C. Does it work, for whom, what can go wrong, and who is accountable?
   - D. Is it the newest?

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